

Mathematical Modeling of Multinuclear Scaling in Muscle Cells

Michelle Lu, Dr. Alex Mogilner

New York University Center for Data Science, New York, NY. 10011, USA

GSTEM 2024

Abstract

In this study, we present a mathematical model and simulation to explore multinuclear scaling in muscle cells, focusing on the dynamic behavior of nuclei in response to signaling from the Neuromuscular Junction (NMJ). The NMJ plays a critical role in muscle function by releasing signaling molecules into the muscle fiber. Our simulation models how these signaling molecules influence the positioning of nuclei within the cell. Specifically, the nuclei adjust their repulsive forces based on the local concentration of signaling molecules: as the concentration of these molecules increases, the repulsive forces between nuclei decrease. This interaction leads to a self-organizing distribution of nuclei, ensuring efficient cellular function. Our findings shed light on the complex relationship between external signaling and intracellular dynamics, providing insights into the mechanisms that regulate nuclear positioning in muscle fibers. The model offers a framework for understanding how disruptions in these processes could contribute to neuromuscular disorders and may inform future therapeutic strategies.

Introduction

The spatial organization of nuclei within muscle cells is a critical aspect of cellular function, particularly in multinucleated muscle fibers where efficient distribution of nuclei is essential for maintaining cellular homeostasis and muscle performance. Multinuclear scaling, or the relative positioning and spacing of multiple nuclei within a single muscle fiber, ensures that each nucleus can effectively manage gene expression, protein synthesis, and other essential processes within its local domain. Disruptions in nuclear positioning have been implicated in various muscle-related diseases, including muscular dystrophies and age-related muscle degeneration, highlighting the importance of understanding the mechanisms underlying nuclear distribution in muscle fibers.

Previous research has explored various factors influencing nuclear positioning, including the role of cytoskeletal elements such as microtubules and actin filaments, as well as molecular motors like dynein and kinesin, which transport nuclei within the cell (Metzger et al., 2012; Roman & Gomes, 2018). Additionally, studies have shown that signaling pathways, such as those involving the Neuromuscular Junction (NMJ), play a crucial role in regulating nuclear dynamics by modulating intracellular forces and signaling cascades (Sanes & Lichtman, 2001; Valdez et al., 2010). However, the precise mechanisms by which external signals from the NMJ influence nuclear positioning and scaling within muscle fibers remain unclear.

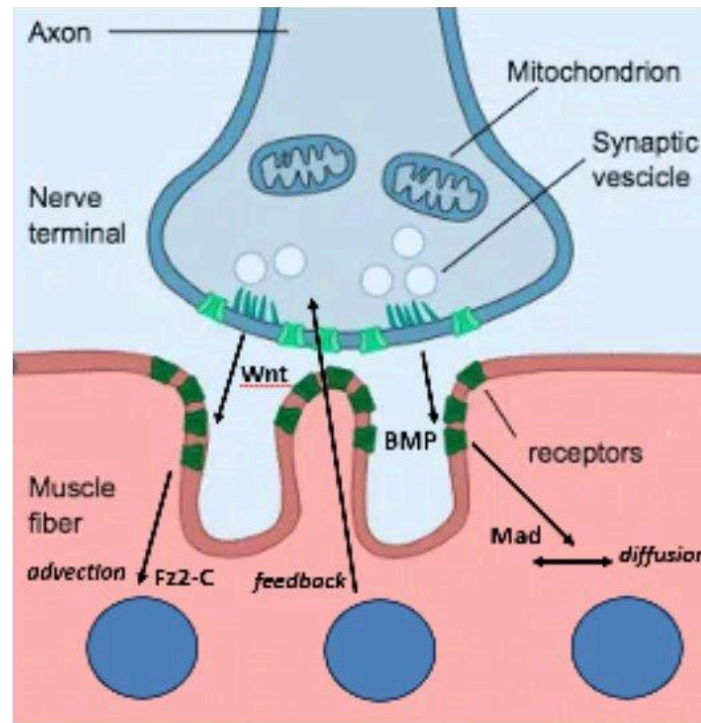


Figure 1: Neuromuscular Junction

This study aims to address this gap by developing a mathematical model and simulation that captures the dynamic behavior of nuclei in response to signaling molecules released by the NMJ. Specifically, our model hypothesizes that the concentration of signaling molecules influences the repulsive forces between nuclei, with higher concentrations leading to a reduction in repulsion and thus facilitating a more compact nuclear arrangement. This hypothesis is grounded in the idea that such a mechanism would allow muscle cells to adapt to varying levels of external signals, optimizing nuclear distribution for different physiological conditions.

The significance of this study lies in its potential to enhance our understanding of the complex interplay between external signaling and intracellular dynamics in muscle cells. By elucidating the mechanisms of multinuclear scaling, this research may provide insights into the etiology of neuromuscular disorders and contribute to the development of therapeutic strategies aimed at correcting nuclear mispositioning in diseased muscle fibers. Furthermore, this study adds to the growing body of literature on cellular organization, offering a novel perspective on how muscle cells maintain their structural and functional integrity in response to external cues.

Methodology

The study was conducted using a simulation model developed in MATLAB R2023a to explore the behavior of nuclei within a muscle cell in response to signaling molecules released by the Neuromuscular Junction (NMJ). The simulation aimed to replicate the dynamic interactions between nuclei and signaling molecules, focusing on how these interactions influence nuclear positioning and spacing within the cell.

Simulation Parameters:

The simulation was designed to model a muscle cell of length 20 arbitrary units containing 10 nuclei. The initial positions of the nuclei were randomly assigned within the cell's length using MATLAB's random number generator. The diffusion of both the nuclei and the signaling molecules was modeled using a set of parameters to simulate their movement over time. The nuclei were given a diffusion coefficient of 0.02 units, reflecting their random motion within the cell, while the signaling molecules were assigned a diffusion coefficient of 2 units, allowing for faster spread throughout the cell.

Signaling Molecule Dynamics:

At the start of the simulation, 20 signaling molecules were randomly distributed between predefined coordinates, simulating their initial release from the NMJ. During each time step of the simulation, 50 new signaling molecules were secreted into the system, with their positions determined randomly within the same predefined region. A degradation probability of 0.01 was applied to each signaling molecule at each time step to account for their natural decay over time. This dynamic process was repeated over 1000 time steps, with each step representing a time increment of 0.1 units.

Nuclei Repulsion and Positioning:

The repulsive forces between the nuclei were modeled as a function of the local concentration of signaling molecules. The repulsion strength for each nucleus was calculated based on the number of nearby signaling molecules within a defined interaction range (L), set to 1 unit. The repulsion strength varied between a minimum of 1 unit and a maximum of 10 units, with higher concentrations of signaling molecules reducing the repulsive force. This adjustment allowed the simulation to model the behavior where increased signaling molecule presence led to a more compact nuclear arrangement.

The positions of the nuclei were updated at each time step based on their calculated repulsion forces and random displacements. Boundary conditions were enforced to ensure that the nuclei and signaling molecules remained within the confines of the cell. The updated positions were stored at each time step, with a particular focus on the final 500 steps to analyze the steady-state behavior of the system.

Data Analysis:

After the simulation was completed, the positions of the nuclei over the final 500 time steps were aggregated and analyzed. The distribution of nuclei along the cell and the spacing between them were calculated. The simulation data were stored in CSV format for further analysis. A linear model was fitted to the data using MATLAB's statistical tools to assess the influence of nuclear position on spacing and to identify any trends in the distribution of nuclei within the cell.

Reproducibility and Validation:

To ensure the robustness of the results, the simulation was run multiple times (10 separate runs), with the results from each run aggregated to provide a comprehensive view of the nuclear distribution patterns. The consistency of these results across multiple runs helped validate the reliability of the simulation model and its ability to replicate the dynamic interactions within a multinucleated muscle cell.

This methodology provides a detailed account of how the simulation was conducted, including the parameters used, the processes modeled, and the methods of data analysis, ensuring that the study can be reproduced and its findings validated by other researchers in the field.

Results

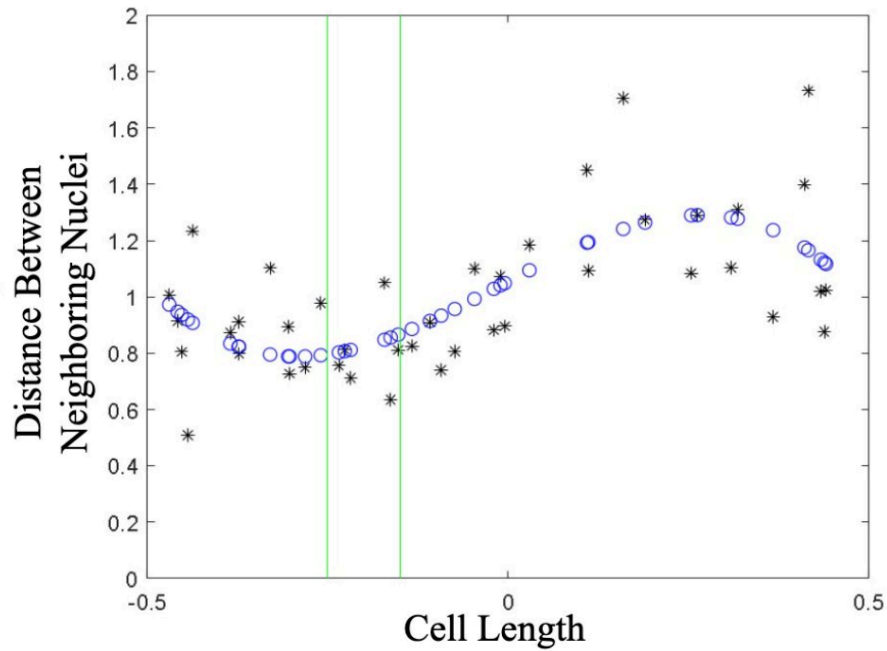


Figure 2: Distance Between Neighboring Nuclei vs. Cell Length

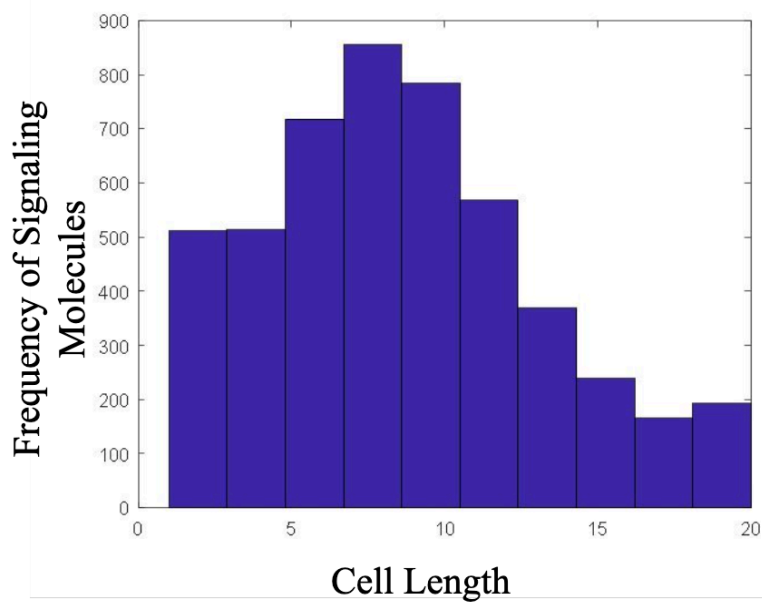


Figure 3: Frequency of Signaling Molecules vs. Cell Length

The simulation results are summarized in the two figures presented below, highlighting the relationship between the distribution of signaling molecules and the spacing between neighboring nuclei within the muscle cell.

Figure 2 illustrates the distance between neighboring nuclei as a function of their normalized position along the cell length. The x-axis represents the cell length, normalized to a range of -0.5 to 0.5, while the y-axis indicates the normalized distance between neighboring nuclei. The black asterisks show the observed distances in the simulation, and the blue circles represent a fitted model. A notable observation from the figure is that in regions where the distance between nuclei decreases, the corresponding positions align with areas where signaling molecules are more concentrated.

Figure 3 shows the frequency distribution of signaling molecules along the cell length. The x-axis represents the cell length (0 to 20 units), while the y-axis shows the frequency of signaling molecules within specified intervals. The histogram reveals a higher concentration of signaling molecules in the central region of the cell, particularly between positions 5 and 10.

When comparing the two figures, it is evident that the decrease in distance between neighboring nuclei observed in **Figure 2** occurs in the same region of the cell where there is a high frequency of signaling molecules, as shown in **Figure 3**. This spatial correlation suggests that the presence of a higher concentration of signaling molecules, which are secreted from the NMJ, corresponds with a tighter clustering of nuclei in that region of the cell and a weaker repulsive force between nuclei.

Discussion

The primary aim of this study was to explore how signaling molecules released from the Neuromuscular Junction (NMJ) influence the positioning and spacing of nuclei within a multinucleated muscle cell. Specifically, the hypothesis posited that an increase in signaling molecules would result in a decrease in the repulsive forces between neighboring nuclei, leading to a more compact arrangement of nuclei in regions with higher signaling molecule concentrations. The data generated by the simulation support this hypothesis, as evidenced by the results presented in Figures 2 and 3.

In **Figure 2**, we observe a clear reduction in the distance between neighboring nuclei in regions of the cell where there is a higher frequency of signaling molecules, as shown in **Figure 3**. This finding aligns with the proposed hypothesis, suggesting that the local concentration of signaling molecules indeed plays a significant role in modulating nuclear positioning. The observed correlation between high concentrations of signaling molecules and tighter nuclear

spacing supports the notion that signaling molecules act to reduce the repulsive forces between nuclei, allowing them to come closer together.

Comparison to Previous Research:

These findings are consistent with previous studies that have highlighted the role of signaling pathways in nuclear positioning within muscle fibers. For example, earlier research by Roman and Gomes (2018) demonstrated that intracellular signaling can influence cytoskeletal dynamics, which in turn affects nuclear distribution. However, our study provides a more direct computational approach to modeling the specific impact of signaling molecule concentration on nuclear spacing. This adds to the existing body of literature by offering a quantitative perspective on how external signals from the NMJ can directly influence intracellular organization.

Modifications and Improvements:

While the results of this study support the initial hypothesis, several aspects of the study design could be refined to enhance the robustness and applicability of the findings. Firstly, the current model assumes a uniform diffusion coefficient for both nuclei and signaling molecules, which may not fully capture the complex dynamics within a real muscle cell. Future iterations of the model could incorporate varying diffusion coefficients or more complex models of intracellular transport mechanisms, such as active transport along microtubules, to more accurately reflect the intracellular environment.

Additionally, the model could be expanded to include other factors that influence nuclear positioning, such as mechanical forces exerted by the surrounding cytoskeleton or interactions with other cellular components. Incorporating these factors could provide a more holistic understanding of the mechanisms driving nuclear distribution in muscle fibers.

Finally, experimental validation of the model's predictions would be a crucial next step. While the simulation results are promising, testing the model's predictions in biological systems would strengthen the conclusions and provide further insights into the role of NMJ signaling in nuclear positioning.

Conclusion

In conclusion, this study provides computational evidence that supports the hypothesis that signaling molecules from the NMJ influence nuclear positioning within muscle cells by reducing repulsive forces between nuclei. The findings are consistent with existing research and offer new quantitative insights into the dynamics of multinuclear scaling. Future work should

focus on refining the model and validating the findings experimentally to further elucidate the mechanisms governing nuclear positioning in muscle cells.

Acknowledgements

Thank you to Dr. Alex Mogilner for his invaluable mentorship and guidance throughout this summer. Additionally, I would like to thank Matthew Leingang and Catherine Tissot for providing me with the opportunity to participate in NYU GSTEM. Finally, thank you to Apoorva Sharma for her support throughout this entire process.

References

1. Metzger, T., Gache, V., Xu, M., Cadot, B., Folker, E., Richardson, B., Gomes, E. R., & Baylies, M. K. (2012). MAP and kinesin-dependent nuclear positioning is required for skeletal muscle function. *Nature*, 484(7392), 120-124.
2. Roman, W., & Gomes, E. R. (2018). Nuclear positioning in skeletal muscle. *Seminars in Cell & Developmental Biology*, 82, 51-56.
3. Sanes, J. R., & Lichtman, J. W. (2001). Induction, assembly, maturation and maintenance of a postsynaptic apparatus. *Nature Reviews Neuroscience*, 2(11), 791-805.
4. Valdez, G., Tapia, J. C., Kang, H., Clemenson, G. D., Gage, F. H., Lightfoot, R.,... & Sanes, J. R. (2010). Attenuation of age-related changes in mouse neuromuscular synapses by caloric restriction and exercise. *Proceedings of the National Academy of Sciences*, 107(4), 1482-1487.
5. Hansson, Kenth-Arne, and Einar Eftestøl. "Scaling of Nuclear Numbers and Their Spatial Arrangement in Skeletal Muscle Cell Size Regulation." Edited by Trina Schroer, *Molecular Biology of the Cell*, 20 June 2023, www.molbiolcell.org/doi/pdf/10.1091/mbc.e17-03-0138.
6. Li, Lei, et al. "Neuromuscular Junction Formation, Aging, and Disorders." *Annual Review of Physiology*, Annual Reviews, 10 Feb. 2018, www.annualreviews.org/content/journals/10.1146/annurev-physiol-022516-034255.
7. Marshall, Wallace F. "Scaling of Subcellular Structures." *Annual Review of Cell and Developmental Biology*, Annual Reviews, 6 Oct. 2020, www.annualreviews.org/content/journals/10.1146/annurev-cellbio-020520-113246.

8. Cadot, B., Gache, V., & Gomes, E. R. (2015). Moving and positioning the nucleus in skeletal muscle – one step at a time. *Nucleus*, 6(5), 373–381.
<https://doi.org/10.1080/19491034.2015.1090073>
9. Manhart A, Windner S, Baylies M, Mogilner A (2018) Mechanical positioning of multiple nuclei in muscle cells. *PLoS Comput Biol* 14(6): e1006208
10. Windner et al., 2019, *Developmental Cell* 49, 48–62 April 8, 2019 ^a 2019 Elsevier Inc.